

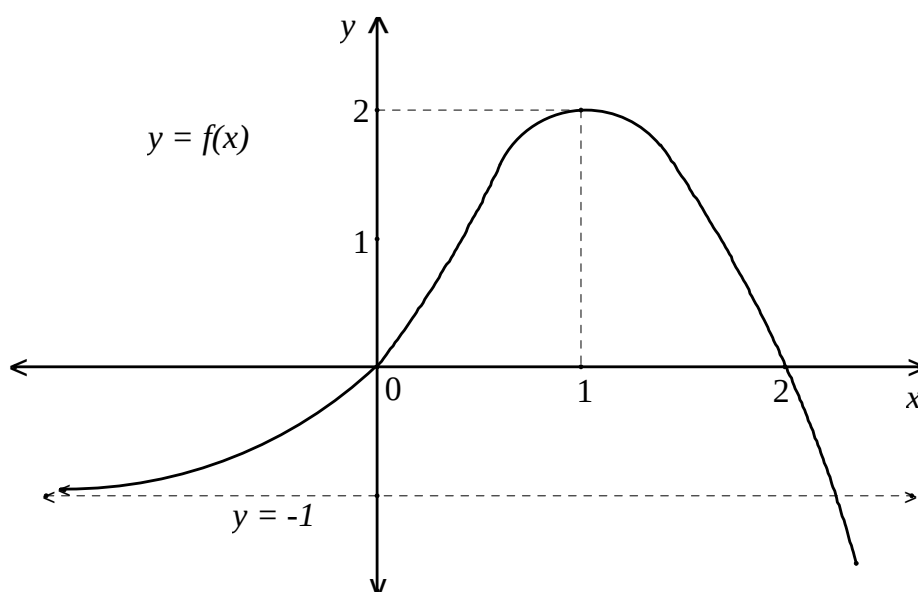
QUESTION 1 (15 Marks)		Marks
(a)	(i) Find: $\int \tan^2 4\theta \, d\theta$.	2
	(ii) Find $\int \frac{dx}{x^2 + 2x + 4}$.	3
	(iii) Evaluate: $\int_0^1 \frac{x}{\sqrt{2-x}} \, dx$.	3
	(iv) Evaluate: $\int_1^3 \frac{4}{x^2 - 4x} \, dx$	3
(b)	Draw a neat sketch of the hyperbola $3x^2 - y^2 = 12$ showing	
	(i) the intercepts with the co-ordinate axes,	1
	(ii) the positions and co-ordinates of the foci,	1
	(iii) the positions and equations of the directrices.	1
	(iv) the positions and equations of the asymptotes.	1

QUESTION 2 (15 Marks) (START A NEW PAGE)		Marks
(a)	(i) Evaluate: $\int_1^e x \ln x \, dx$.	3
	(ii) Evaluate: $\int_0^{\frac{\pi}{4}} \sin^3 \theta \, d\theta$.	3
	(iii) Using the substitution $u = e^x$, find $\int \frac{e^{2x}}{e^x - 1} \, dx$.	3
(b)	(i) Draw a neat sketch of the ellipse $9x^2 + 25y^2 = 225$ showing	
	(α) the intercepts with the co-ordinate axes,	1
	(β) the positions and co-ordinates of the foci,	1
	(γ) the positions and equations of the directrices.	1
	(ii) Write down the equation of the tangent to the ellipse at the point $P(5\cos\theta, 3\sin\theta)$.	1
	(iii) The tangent at P meets the tangents from the ends of the major axis at the points Q and R . Prove that the intervals QS and RS are perpendicular where S is the focus with the positive x -coordinate.	2

- (a) (i) Express $\frac{1}{1+x+x^2+x^3}$ in the form $\frac{A}{1+x} + \frac{Bx+C}{1+x^2}$, where A , B and C are rational numbers. 2

- (ii) Hence evaluate $\int_0^1 \frac{dx}{1+x+x^2+x^3}$. 3

- (b) The graph of $y = f(x)$ is illustrated. The line $y = -1$ is a horizontal asymptote.



Using the separate graphs provided, sketch each of the graphs below. In each case, clearly label any maxima or minima and the equations of any asymptotes.

- (i) $y = \frac{1}{f(x)}$. 2
- (ii) $y = 2f(x+1)$. 2
- (c) (i) Given the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, prove that the line $y = mx + c$ is a tangent provided that $c^2 = a^2m^2 + b^2$. 2
- (ii) Two tangents are drawn from the point $T(p, q)$ to touch the above ellipse at points M and N . Show that the gradients of these tangents are the roots of the quadratic equation $(a^2 - p^2)m^2 + 2pqm + (b^2 - q^2) = 0$. 3
- (iii) If the above two tangents are perpendicular, show that $a^2 + b^2 = p^2 + q^2$. 1

(a) (i) Given that $I_n = \int_0^{\frac{\pi}{2}} \theta \sin^n \theta \, d\theta$, prove that $I_n = \frac{n-1}{n} I_{n-2} + \frac{1}{n^2}$ for $n \geq 2$. **3**

(ii) Hence show that $\int_0^{\frac{\pi}{2}} \theta \sin^5 \theta \, d\theta = \frac{149}{225}$. **3**

(b) (i) Use the substitution $t = \tan \frac{x}{2}$ to prove that $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x} = \frac{\pi}{3\sqrt{3}}$. **3**

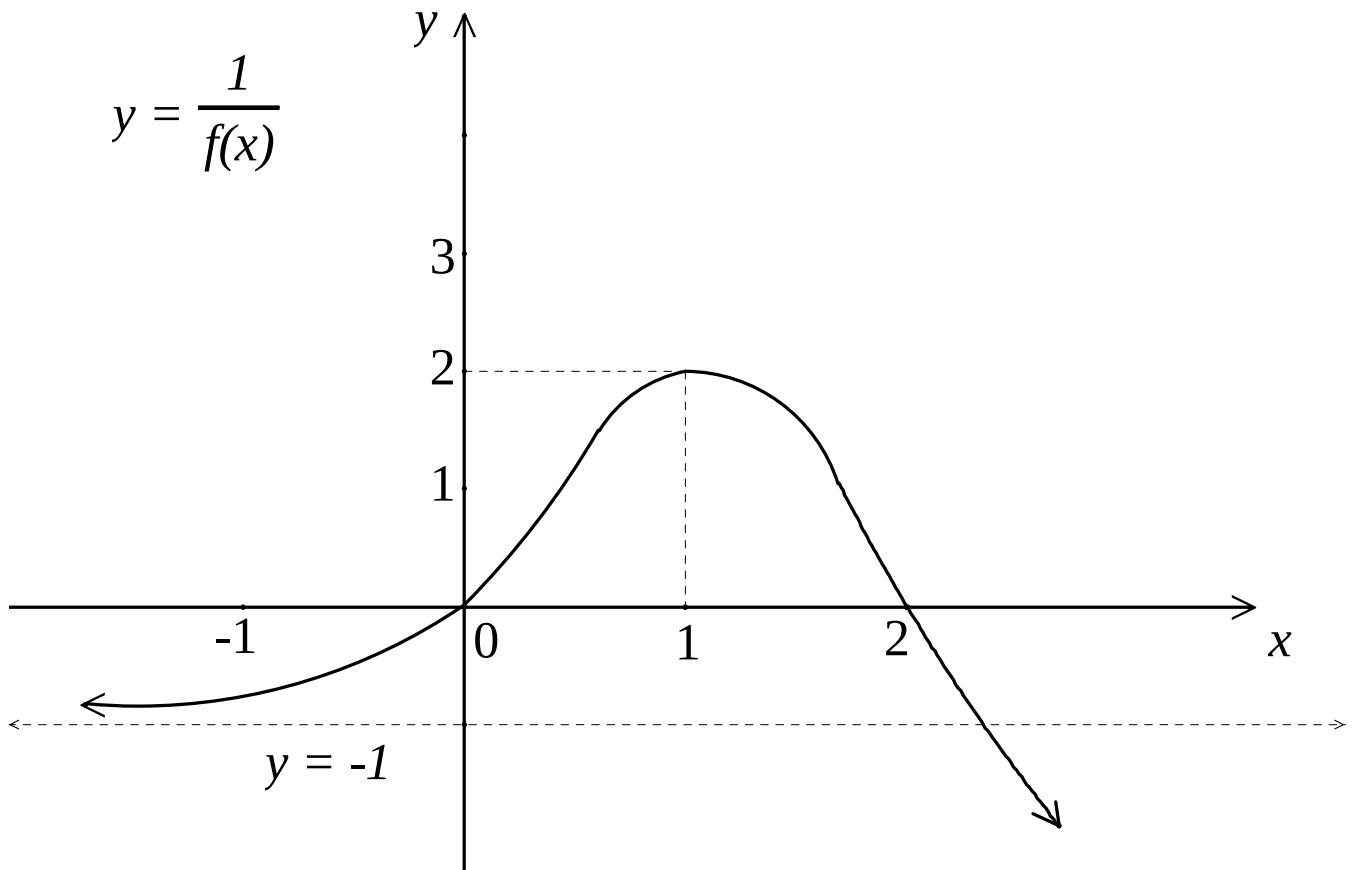
(ii) Show that $\int_0^{2a} f(x) \, dx = \int_0^a \{f(x) + f(2a-x)\} \, dx$. **3**

(iii) Hence evaluate $\int_0^{\pi} \frac{x}{2 + \sin x} \, dx$. **3**

►► THIS IS THE END OF THE EXAMINATION PAPER ◀◀

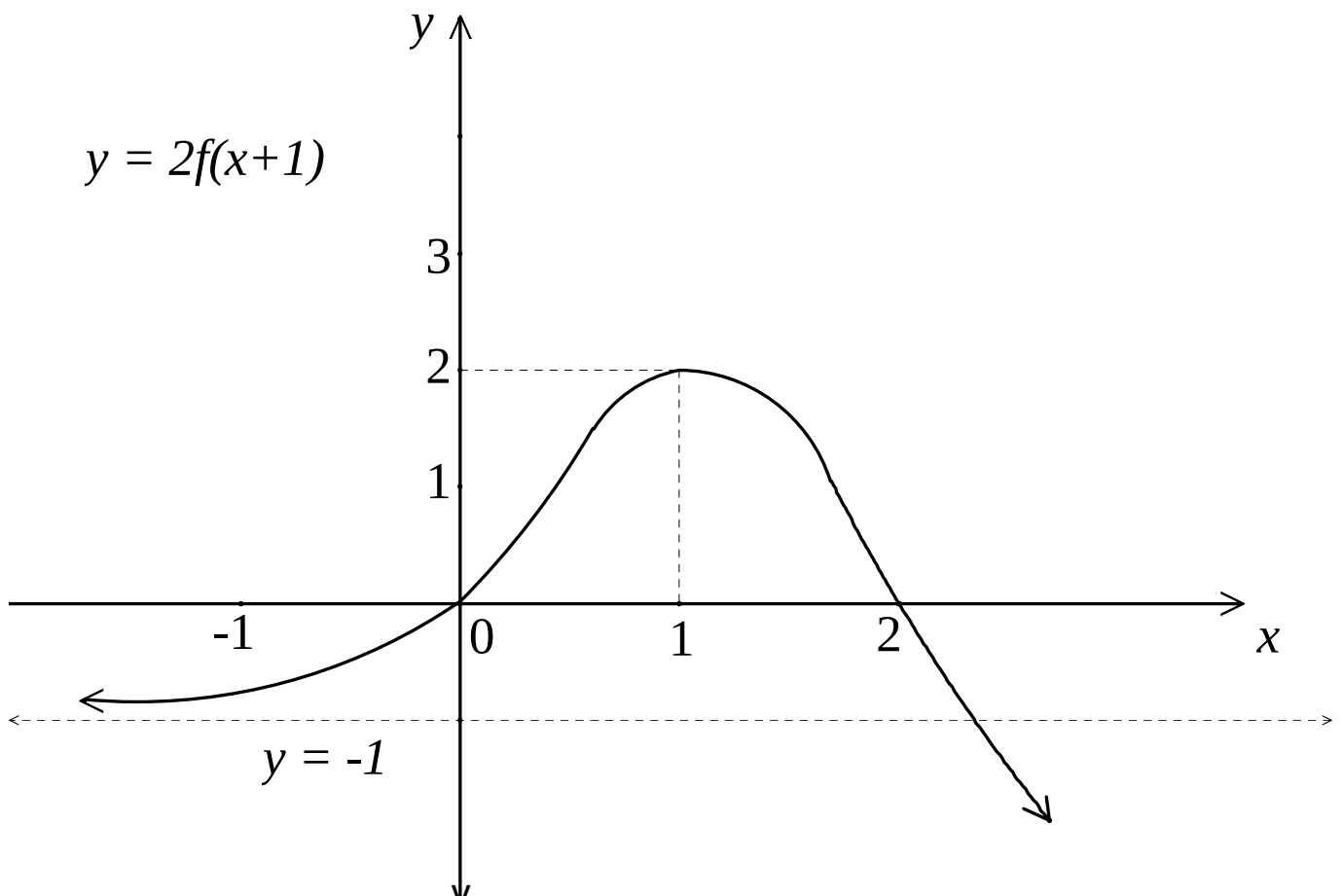
Question 3(b)(i)

Student #:



Question 3(b)(ii)

Student #:



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MATHEMATICS Extension 1 : Question..... (p2 of 2)

MATHEMATICS Extension 1 : Question 1.... (p2 of 2)		
Suggested Solutions	Marks	Marker's Comments
$= -\int_1^3 \frac{dx}{x} + \int_1^3 \left(\frac{-dx}{4-x} \right) \quad (\text{Ensuring correct domain})$ $= \left[-\ln x + \ln(4-x) \right]_1^3$ $= -\ln 3 + \ln 1 + \ln 1 - \ln 3$ $= \underline{\underline{-2\ln 3}}$	1	Many used 11 signs. Any mention of $\ln(-1)$ or $\ln(-3)$ lost a mark.
b) $\frac{x^2}{4} - \frac{y^2}{12} = 1 \quad \therefore a^2 = 4, b^2 = 12$ $b^2 = a^2(e^2 - 1) \Rightarrow e^2 = 4$ $\Rightarrow \underline{\underline{e = 2}} \quad (e > 0)$ Foci: $(\pm ae, 0) = \underline{\underline{(\pm 4, 0)}}$ Directrices: $x = \pm \frac{a}{e} \rightarrow \underline{\underline{x = \pm 1}}$ Intercepts: $(y=0) \quad x = \pm 2 \rightarrow \underline{\underline{(\pm 2, 0)}}$ $(x=0) \quad \text{No such points}$ Asymptotes $\underline{\underline{y = \pm x\sqrt{3}}}$	4	generally well done. 1/2 mark deducted if $y = x\sqrt{3}$ was not above 45°

Y12 MATH EXT2 ASSESSMENT TASK 2

TERM 1, 2011

MATHEMATICS Extension 2: Question 2

Suggested Solutions	Marks	Marker's Comments
Q2(a) (i) $I = \int_1^e x \ln x \, dx$ $u = \ln x \quad dv = x \, dx$ $du = \frac{1}{x} \, dx \quad v = \frac{1}{2} x^2$ $I = \left[\frac{1}{2} x^2 \ln x \right]_1^e - \frac{1}{2} \int_1^e x \, dx$ ✓ $= \left[\frac{1}{2} e^2 - 0 \right] - \frac{1}{2} \times \frac{1}{2} x^2 \Big _1^e$ $= \frac{1}{2} e^2 - \frac{1}{4} (e^2 - 1)$ ✓ $= \frac{1}{2} e^2 - \frac{1}{4} e^2 + \frac{1}{4}$ $= \frac{1}{4} (e^2 + 1)$	$\frac{1}{2}$ 1	1 For correct IBPs $\frac{1}{2}$ for $\frac{1}{2} e^2 - 0$ [3]
(ii) $I = \int_0^{\pi/4} \sin^3 \theta \, d\theta$ $= \int_0^{\pi/4} \sin \theta \times \sin^2 \theta \, d\theta =$ $= \int_0^{\pi/4} \sin \theta (1 - \cos^2 \theta) \, d\theta$ ✓ $= \int_0^{\pi/4} \sin \theta - \sin \theta \cdot \cos^2 \theta \, d\theta$ $= \left[-\cos \theta + \frac{1}{3} \cos^3 \theta \right]_0^{\pi/4}$ ✓ $= \left[\frac{1}{3} \cos^3 \theta - \cos \theta \right]_0^{\pi/4}$ $= \left(\frac{1}{3} \cdot \frac{1}{2\sqrt{2}} - \frac{1}{\sqrt{2}} \right) - \left(\frac{1}{3} - 1 \right)$ $= \frac{1}{6\sqrt{2}} - \frac{1}{\sqrt{2}} - \left(-\frac{2}{3} \right)$ $= \frac{2}{3} - \frac{5}{6\sqrt{2}} = \frac{4\sqrt{2} - 5}{6\sqrt{2}} = \frac{8 - 5\sqrt{2}}{12}$	$\frac{1}{2}$ 1 $\frac{1}{2}, \frac{1}{2}$ $\frac{1}{2}$	Method 2 Let $u = \cos \theta$ $du = -\sin \theta \, d\theta$ $\sin \theta \, d\theta = -du$ $\theta \quad u$ $0 \quad \frac{1}{\sqrt{2}}$ $\frac{\pi}{4} \quad \frac{1}{\sqrt{2}}$ $I = \int_{\frac{1}{\sqrt{2}}}^1 (u^2 - 1) \, du$ etc Method 3 IBPs $u = \sin^2 \theta \quad dv = \sin \theta \, d\theta$ $du = 2 \sin \theta \cos \theta \, d\theta$ $v = -\cos \theta$ $I = \left[-\cos \theta \sin^2 \theta \right]_0^{\pi/4} + 2 \int_0^{\pi/4} \sin \theta \cos^3 \theta \, d\theta$ $= -\frac{1}{2\sqrt{2}} - 0 - \frac{2}{3} \cos^3 \theta \Big _0^{\pi/4}$ [3]
(iii) $I = \int \frac{e^{2x}}{e^x - 1} \, dx$ $u = e^x \quad [x = \ln u]$ $du = e^x \, dx = u \, dx$ $\therefore dx = \frac{du}{u}$ $I = \int \frac{u^2}{u-1} \times \frac{du}{u} = \int \frac{u \, du}{u-1}$ $= \int 1 + \frac{1}{u-1} \, du$ $= u + \ln u-1 + C$ $\therefore I = e^x + \ln e^x - 1 + C$	✓ ✓ ✓	[3] $-\frac{1}{2}$ if no Abs. brackets

Solutions to Ext 2 T1 Q3

a) $A = \frac{1}{2}$

$B = -\frac{1}{2}$

$C = \frac{1}{2}$

$$\frac{1}{2(1+x)} + \frac{-x+1}{2(1+x^2)}$$

$$\begin{aligned} \text{ii) } \int_0^1 \frac{dx}{2(1+x)} - \int_0^1 \frac{x dx}{2(1+x^2)} + \int_0^1 \frac{dx}{2(1+x^2)} \\ = \frac{1}{2} \left[\ln(1+x) \right]_0^1 - \frac{1}{4} \left[\ln(1+x^2) \right]_0^1 + \frac{1}{2} \left[\tan^{-1} x \right]_0^1 \end{aligned}$$

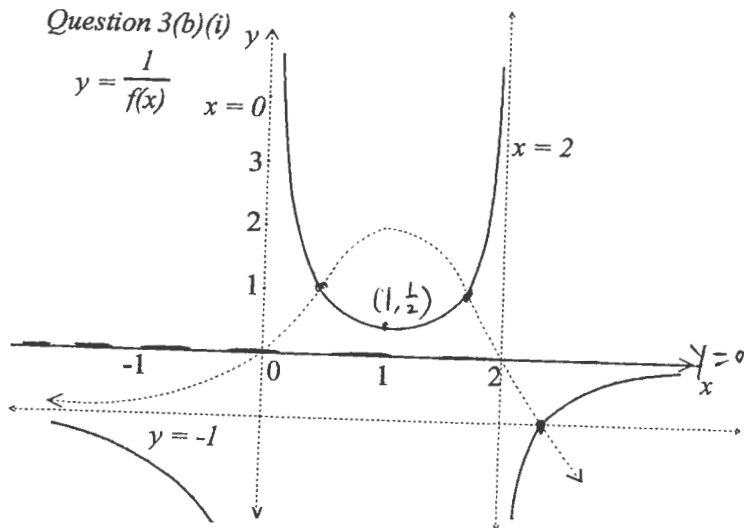
$$= \frac{1}{2} \ln 2 - \frac{1}{4} \ln 2 - \frac{1}{2} \frac{\pi}{4}$$

$$= \frac{\ln 2}{4} + \frac{\pi}{8}$$

b)

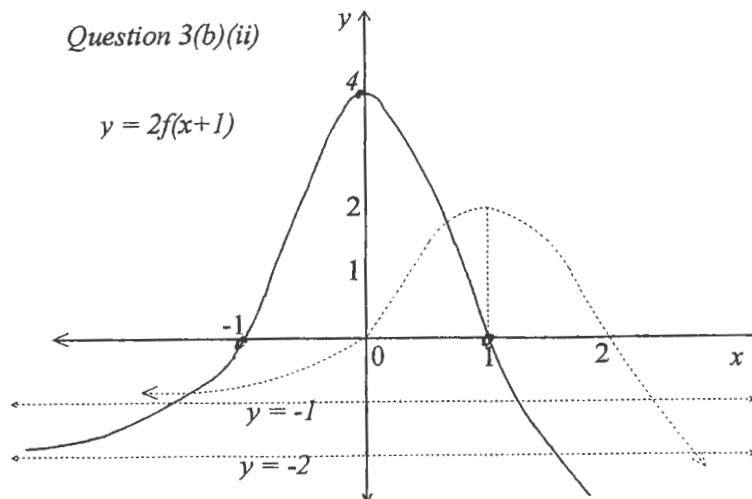
Question 3(b)(i)

$$y = \frac{1}{f(x)} \quad x=0$$



Question 3(b)(ii)

$$y = 2f(x+1)$$



$$\frac{1}{2} m$$

$$\frac{1}{2} m$$

$$\frac{1}{2} m$$

$$\frac{1}{2} m$$

Some forgot this line

$$\frac{1}{2} m + \frac{1}{2} m + \frac{1}{2} m$$

$$1 m$$

$$\frac{1}{2} m$$

Asy $x=0, x=2, y=0, y=1$ $\frac{1}{2} m$
Each branch $\frac{1}{2} m @ \Rightarrow \frac{1}{2} m$
must meet $y=1, -1$
must show min $(1, \frac{1}{2})$

Asy $y=-2$ $\frac{1}{2} m$
 x, y intercepts $\frac{1}{2} m$
Each side graph $\frac{1}{2} m \Rightarrow 1 m$

$$(i) \frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$$

$\frac{1}{2}m$

$$(b^2 + a^2m^2)x^2 + 2a^2mcx + a^2(c^2 - b^2) = 0$$

$\frac{1}{2}m$

tangent if $\Delta = 0$

$$4a^2m^2c^2 - 4(b^2 + a^2m^2) \times a^2(c^2 - b^2) = 0$$

$\frac{1}{2}m$

$$4a^2b^2(a^2m^2 - c^2 + b^2) = 0$$

$\frac{1}{2}m$

$$a \neq 0, b \neq 0$$

$$c^2 = a^2m^2 + b^2$$

$$(ii) \quad q = mp + c$$

1 m

$$c^2 = a^2m^2 + b^2$$

$$(q - mp)^2 = a^2m^2 + b^2$$

1 m

$$q^2 - 2mpq + m^2p^2 = a^2m^2 + b^2$$

$$(a^2 - p^2)m^2 + 2pqm + (b^2 - q^2) = 0$$

1 m

$$(iii) \quad m_1 \times m_2 = -1 \quad \text{for perpendicular lines}$$

$$\frac{b^2 - q^2}{a^2 - p^2} = -1$$

product of roots is c/a

1 m

$$a^2 + b^2 = p^2 + q^2$$

Some of students write:
 $c = \sqrt{a^2m^2 + b^2}$
 $(-1m)$

- (a) (i) Given that $I_n = \int_0^{\frac{\pi}{2}} \theta \sin^n \theta d\theta$, prove that $I_n = \frac{n-1}{n} I_{n-2} + \frac{1}{n^2}$ for $n \geq 2$.

3

Solution:

$$\begin{aligned}
 I_n &= \int_0^{\frac{\pi}{2}} \theta \sin^n \theta d\theta \\
 &= \int_0^{\frac{\pi}{2}} \left\{ \theta \sin^{n-1} \theta \times \frac{d}{d\theta} (-\cos \theta) \right\} d\theta \\
 &= uv - \int v du \\
 &= \left[-\theta \cos \theta \sin^{n-1} \theta \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \left\{ (-\cos \theta) \times (\sin^{n-1} \theta + \theta(n-1) \cos \theta \sin^{n-2} \theta) \right\} d\theta \\
 &= 0 + \int_0^{\frac{\pi}{2}} (\cos \theta \sin^{n-1} \theta + (n-1) \theta \cos^2 \theta \sin^{n-2} \theta) d\theta \\
 &= \int_0^{\frac{\pi}{2}} (\cos \theta \sin^{n-1} \theta) d\theta + (n-1) \int_0^{\frac{\pi}{2}} (\theta \cos^2 \theta \sin^{n-2} \theta) d\theta \\
 &= \left[\frac{1}{n} \sin^n \theta \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \{ \theta (1 - \sin^2 \theta) \sin^{n-2} \theta \} d\theta \\
 &= \frac{1}{n} + (n-1) \int_0^{\frac{\pi}{2}} \{ \theta \sin^{n-2} \theta - \theta \sin^n \theta \} d\theta \\
 &= \frac{1}{n} + (n-1) \int_0^{\frac{\pi}{2}} \{ \theta \sin^{n-2} \theta \} d\theta - (n-1) \int_0^{\frac{\pi}{2}} \{ \theta \sin^n \theta \} d\theta \\
 &= \frac{1}{n} + (n-1) I_{n-2} - (n-1) I_n \\
 I_n + (n-1) I_n &= \frac{1}{n} + (n-1) I_{n-2} \\
 n I_n &= \frac{1}{n} + (n-1) I_{n-2} \\
 I_n &= \frac{1}{n^2} + \frac{n-1}{n} I_{n-2}
 \end{aligned}$$

- (ii) Hence show that $\int_0^{\frac{\pi}{2}} \theta \sin^5 \theta d\theta = \frac{149}{225}$.

3

Solution:

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \theta \sin^5 \theta d\theta &= I_5 \\
 I_5 &= \frac{1}{25} + \frac{4}{5} I_3 \\
 I_3 &= \frac{1}{9} + \frac{2}{3} I_1
 \end{aligned}$$

$$\begin{aligned}
 I_1 &= \int_0^{\frac{\pi}{2}} \theta \sin \theta \, d\theta \\
 &= \int_0^{\frac{\pi}{2}} \theta \frac{d}{d\theta} (-\cos \theta) \, d\theta \\
 &= \left[-\theta \cos \theta \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (-\cos \theta) \, d\theta \\
 &= 0 + \int_0^{\frac{\pi}{2}} \cos \theta \, d\theta \\
 &= \left[\sin \theta \right]_0^{\frac{\pi}{2}} \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 I_3 &= \frac{1}{9} + \frac{2}{3}(1) \\
 &= \frac{7}{9}
 \end{aligned}$$

$$\begin{aligned}
 I_5 &= \frac{1}{25} + \frac{4}{5}\left(\frac{7}{9}\right) \\
 &= \frac{149}{225}
 \end{aligned}$$

$$\int_0^{\frac{\pi}{2}} \theta \sin^5 \theta \, d\theta = \frac{149}{225}$$

- (b) (i) Use the substitution $t = \tan \frac{x}{2}$ to prove that $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x} = \frac{\pi}{3\sqrt{3}}$.

3

Solution:

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x} &= \int_0^1 \left(\frac{1}{2 + \frac{2t}{1+t^2}} \right) \times \frac{2dt}{1+t^2} \\
 &= \int_0^1 \left(\frac{2dt}{2(1+t^2) + 2t} \right) \\
 &= \int_0^1 \left(\frac{1}{t^2 + t + 1} \right) dt \\
 &= \int_0^1 \left(\frac{1}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}} \right) dt
 \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{\sqrt{3}} \left[\tan^{-1} \left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) \right]_0^1 \\
&= \frac{2}{\sqrt{3}} \left[\tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) \right]_0^1 \\
&= \frac{2}{\sqrt{3}} \left\{ \tan^{-1} \left(\frac{3}{\sqrt{3}} \right) - \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \right\} \\
&= \frac{2}{\sqrt{3}} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) \\
&= \frac{\pi}{3\sqrt{3}}
\end{aligned}$$

(ii) Show that $\int_0^{2a} f(x) dx = \int_0^a \{f(x) + f(2a-x)\} dx$.

3

Solution:

$$\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_a^{2a} f(x) dx$$

Consider

$$\int_a^{2a} f(x) dx, \text{ let } u = 2a - x, du = -dx, x = a \Rightarrow u = a, x = 2a \Rightarrow u = 0$$

$$\int_a^{2a} f(x) dx = \int_a^0 f(2a-u) (-du)$$

$$= \int_0^a f(2a-u) du$$

$$= \int_0^a f(2a-x) dx \text{ (let } u = x)$$

$$\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx$$

$$= \int_0^a \{f(x) + f(2a-x)\} dx$$

(iii) Hence evaluate $\int_0^{\pi} \frac{x}{2 + \sin x} dx$.

Solution:

$$\begin{aligned}
\int_0^{\pi} \frac{x}{2 + \sin x} dx &= \int_0^{\frac{\pi}{2}} \left(\frac{x}{2 + \sin x} + \frac{\pi - x}{2 + \sin(\pi - x)} \right) dx \\
&= \int_0^{\frac{\pi}{2}} \left(\frac{x}{2 + \sin x} + \frac{\pi - x}{2 + \sin x} \right) dx \\
&= \int_0^{\frac{\pi}{2}} \left(\frac{x + \pi - x}{2 + \sin x} \right) dx
\end{aligned}$$

Alternative

$$\begin{aligned}
\text{LHS} &= \int_0^{2a} f(x) dx = F(x) \Big|_0^{2a} = F(2a) - F(0), \text{ where } F(x) \text{ is a primitive of } f(x). \\
\text{RHS} &= \int_0^a f(x) dx + \int_a^{2a} f(x) dx \\
&= [F(x) - F(2a-x)]_0^a \\
&= F(a) - F(a) - (F(0) - F(2a)) \\
&= F(2a) - F(0) \\
\therefore \text{LHS} &= \text{RHS} \\
\therefore \int_0^{2a} f(x) dx &= \int_0^a f(x) + f(2a-x) dx.
\end{aligned}$$

3